

# Ayman Mahfuz

Software Engineer | Machine Learning Systems

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## EDUCATION

### The University of Texas at Austin

Austin, TX

Bachelor of Science in Computer Science

Aug 2023 – May 2027

- **Concentration:** AI and ML — **GPA:** 3.5/4.0
- **Courses:** HPC, NLP, Generative Visual Computing, Energy-Efficient Computing

## EXPERIENCE

### Arm

May 2025 – Present

Platform Validation Intern | Silicon Bring-Up & ML Infrastructure

- Owned Arm AGI server CPU stress testing and performance characterization across tape-out readiness, **first-silicon bring-up**, and **post-silicon validation**; profiled ML workloads with **PMU counters**.
- Introduced **Bayesian optimization** for stress-test selection, using Gaussian-process surrogates and active-search acquisition functions to identify repeatable, high-yield configurations that exposed novel CPU/memory failures.
- Cut validation test executions **~100x** to **under 1%** of the original suite while preserving **99.8th-percentile failure coverage**; presented to SVP/director leadership and received patent-filing approval.
- Architected modular **Python** search infrastructure with pluggable surrogate-model, acquisition-function, and kernel APIs, enabling new ML and validation strategies without core-pipeline changes.

### Texas Robotics, UTCS AI Lab

Jan 2025 – Present

Undergraduate Research Assistant

- Built hierarchical multi-agent **RL** policies for dribble/pass/shoot/defend behaviors in UT Austin Villa's **400K+ LOC C++/Python** robot soccer simulator, deploying policies on NAO robots in **7v7** matches.
- Optimized curriculum-based **RL** workflows with reward shaping, reproducible GPU configs, and CI-backed experiment setup, cutting training iteration time **2-3x** and scaling runs to **5M+ episodes**.
- Helped UT Austin Villa place **3rd globally** at RoboCup 2025 SPL.

### UT Austin - Center for Media Engagement

Aug 2023 – May 2025

Software Engineer, Undergraduate Research Assistant

- Built and maintained ingestion pipelines (sitemaps, Selenium/HTML scraping, REST collectors) feeding a **250M+** row **BigQuery** corpus of **50M+** articles and **70M+** comments; added cleaning/normalization for downstream **ML** training and evaluation.
- Fine-tuned **BERT** classifiers (clickbait, entity, sentiment, story ID) to **99%** accuracy end-to-end, from dataset construction through validation and error analysis.
- Built a research data-collection platform with three **JavaScript** games, one backend, round-robin assignment, action deduplication, and structured per-user event logging for behavioral analysis, serving **1,000+** participants.

### UT Austin - Oden Institute

Aug 2023 – Jan 2025

Undergraduate Research Assistant

- Engineered a containerized Apptainer/SLURM pipeline on TACC H100s for 3D-MRI segmentation, benchmarking CNNs, vision transformers, and hybrids across **1,000+** scans with **5-fold** cross-validation; **+12%** Dice.
- Resolved GPU memory bottlenecks, I/O lag, and mixed-precision instability for stable large-scale training.

## PROJECTS

### HelmPM (helmpm.app) – LLM-Native PM Platform | React, FastAPI, PostgreSQL, OpenAI/Anthropic

- Launched a multi-tenant **LLM-native PM** platform serving **dozens of users** across **40+ organizations** tracking **thousands of tickets**, with retrieval over docs, tickets, and codebases plus automated Linear/Jira sync.
- Engineered **hybrid retrieval** combining dense **semantic search** and sparse keyword matching for meaning/exact-ID ticket resolution; added content-hash Merkle-tree indexing to re-embed only changed documents.

### The Modality-Dependent Energy Playbook – First-Author Research Paper | Python, PyTorch, vLLM, SLURM, TACC GH200

- Authored a controlled cross-modality inference-energy study (**577 runs, 55 configs, 9 GH200 models**) with a YAML/SLURM harness capturing NVML/Zeus energy under thermal gating and seed control; validated with Welch's t-tests and Bonferroni correction.
- Measured **FP8/INT8** cutting **LLM energy 24-32%** with **zero diffusion savings**, explaining the divergence via roofline analysis (memory-bound decode vs. compute-bound denoising) and refuting cross-modality transfer of energy optimizations.

### Modern Transformer LLM with RLHF Alignment | PyTorch, HPC

- Built a **253M-parameter** decoder-only **LLM** from scratch with RoPE/RMSNorm/SwiGLU/attention sinks; pretrained on **600M tokens** and implemented **SFT/DPO** alignment.
- Implemented the full training stack – training loop, mixed precision, gradient accumulation, checkpoint/resume, and eval harnesses – to keep multi-session pretraining stable and reproducible on HPC.

## SKILLS

**Systems/Performance:** Python, C++, profiling, SLURM, Docker, Linux, vLLM, ZeusMonitor/NVML, TACC/HPC

**ML/AI:** PyTorch, Transformers, LoRA/PEFT, LLMs, Reinforcement Learning (PPO, MAPPO), RLHF/DPO

**Web/Data:** TypeScript, React, FastAPI, Flask, PostgreSQL, BigQuery, Pandas, NumPy, REST APIs, AWS, GCP